



LoanHub Vodacom M-Pesa OpenAPI

Sandbox Testing, Verification and Go-Live Manual

Web release: LoanHub v1.2 M-Pesa Integration
Market: Vodacom Lesotho (vodacomLES / LES / LSL)
Prepared for: Matsepo and LoanHub participating lending companies
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Release statement

This manual validates the implemented LoanHub integration and guides sandbox testing. It does not prove production approval, settlement, or certification. Live operation requires valid Vodacom credentials, product enablement, organisation linking, end-to-end testing, review and go-live approval.

1. Document control and release scope

Item	Value
Document owner	Ithute Solutions / LoanHub platform owner
Version	1.0 testing manual for LoanHub M-Pesa web integration
Date	2026-07-17
Database migration head	b8d5e21f9a30
Backend	FastAPI, SQLAlchemy, Alembic, PostgreSQL, Redis
Frontend	Next.js 16, React 19, TypeScript, Tailwind CSS
M-Pesa market	vodacomLES
Country and currency	LES / LSL
Supported actors	Borrower, company finance/management, platform owner

Purpose

This manual provides repeatable test procedures for LoanHub's M-Pesa payment, disbursement, direct-debit, reversal, monitoring, receipt and support-query features. It is designed for developers, finance testers, company owners, support teams and the platform owner.

What is implemented

Per-company encrypted M-Pesa configuration and SessionKey testing.
 C2B single-stage and multi-stage borrower repayments.
 Query Beneficiary Name before B2C loan disbursement.
 B2C disbursement, B2B business payment, transaction query and status update.
 Direct Debit Create, Payment, Query and Cancel.
 Reversal request, management approval and compensating accounting/loan effects.
 Idempotent callbacks, automatic status reconciliation and PDF transaction documents.
 Platform-owner monitoring, company/borrower query centre and one non-overlapping floating tool dock.

Verification boundary

Important

The exact field names and response fields for each product must be compared with the current Vodacom Lesotho portal page attached to the actual application. The Query Transaction Status contract is confirmed from supplied documentation. Other payloads follow the portal conventions but must be validated during sandbox certification.

2. Manual contents

Section	Topic
3	Architecture and role controls
4	Environment preparation and installation
5	Company M-Pesa configuration
6	SessionKey and connection tests
7	Borrower C2B single-stage repayment
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Test result convention

Result	Meaning
PASS	Observed result matches the expected result.
FAIL	Observed result differs; record evidence and stop go-live for affected feature.
BLOCKED	External credentials, product scope, test number or provider availability prevents execution.
NOT APPLICABLE	The company contract does not enable the product.

3. Architecture and role controls

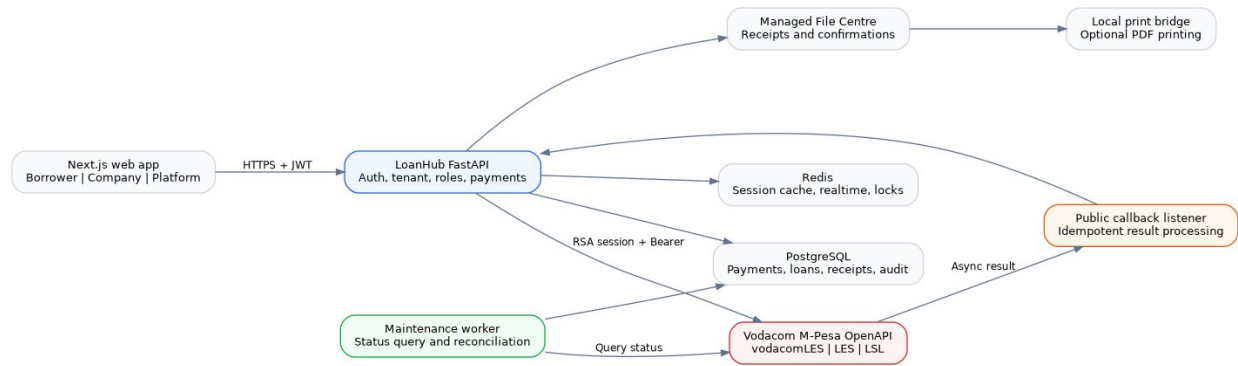


Figure 1 - LoanHub M-Pesa runtime architecture

M-Pesa secrets remain in the backend. Next.js sends normal LoanHub requests authenticated with the user access token and company header. The backend resolves tenant and branch access before calling M-Pesa.

Role matrix

Action	Borrower	Finance role	Company management	Platform owner
View own M-Pesa transactions	Yes	Company/branch scope	Company scope	All companies
Initiate own C2B repayment	Yes	Assisted only when permitted	View/oversight	View/oversight
Configure M-Pesa credentials	No	View/test where allowed	Create/update/enable	Configuration metadata only
Query beneficiary and B2C disburse	No	Yes	Yes	Oversight
B2B payment	No	Company-wide finance only	Yes	Oversight
Create/cancel own direct debit	Yes	Yes for company clients	Yes	Oversight
Collect direct debit	No	Yes	Yes	Oversight
Request reversal	No	Yes	Yes	Oversight
Approve reversal	No	No	Yes	Oversight
Respond to platform queries	Submit	Submit	Submit	Review/respond/status

Branch protection

Branch-scoped staff see only M-Pesa records linked to loans in their assigned branch. Company-level B2B payments require company-wide finance access.

4. Environment preparation and installation

4.1 Prerequisites

PostgreSQL backup and tested restore procedure.

Redis available for session cache, realtime communication and locks.

HTTPS-capable public callback URL for end-to-end sandbox tests.

Vodacom developer application linked to the organisation and required products enabled.

Sandbox API key, public key, service-provider code and trusted origin.

Sandbox MSISDNs/scenarios supplied by Vodacom.

4.2 Back up the database

```
pg_dump --format=custom --no-owner --no-privileges \
"$DATABASE_URL" > loanhub-before-mpesa.dump
```

4.3 Apply the migration

```
cd apps/loan_backend
source .venv/bin/activate
alembic current
alembic heads
alembic upgrade head
alembic current
```

Expected head
b8d5e21f9a30

4.4 Runtime settings

```
ENVIRONMENT=development
PAYMENT MOCK MODE=false
MPESA_HOST=https://openapi.m-pesa.com
MPESA_HTTP_TIMEOUT_SECONDS=45
MPESA_CALLBACK_TOKEN=<random-private-value>
REDIS_URL=redis://localhost:6379/0
```

Company API keys, public keys and shortcodes are entered through the Company M-Pesa page and are stored encrypted. They are not placed in browser environment variables.

4.5 Start services

```
uvicorn main:app --host 0.0.0.0 --port 8000 --reload
python -m docker.maintenance

cd apps/frontend
pnpm install
pnpm typecheck
pnpm lint
pnpm build
pnpm dev
```

5. Company M-Pesa configuration

Open Company > M-Pesa. Only authorised company-management users may save production-sensitive configuration.

1. Select Sandbox for all initial testing.
2. Confirm the market is vodacomLES, country LES and currency LSL.
3. Enter the service-provider code/shortcode assigned to the organisation.
4. Enter the M-Pesa API key and current RSA public key.
5. Enter the exact trusted Origin registered in the developer application.
6. Enter an HTTPS callback base URL for end-to-end callback tests.
7. Enable only the M-Pesa products approved for this application.
8. Keep Production enabled off until Vodacom and the organisation approve go-live.
9. Save the configuration, then run Test connection.

Field	Validation
Environment	sandbox initially; openapi only after explicit production enablement
Market/Country/Currency	vodacomLES / LES / LSL; fixed by LoanHub
Service provider code	4-12 characters according to provider contract
Origin	Must match trusted source; no wildcard in production
Callback base URL	HTTPS for production; points to reachable LoanHub API
API key/Public key	Required on first save; never returned after save
Enabled products	Must match products enabled in the M-Pesa application

6. Generate SessionKey and connection tests

LoanHub encrypts the API key with the configured M-Pesa RSA public key, calls Generate SessionKey and caches the returned session per company and environment.

Test ID	Procedure	Expected result
SES-01	Save valid sandbox credentials and select Test connection.	Success message; last test status and time updated.
SES-02	Use an invalid API key.	Safe failure message; no key appears in response or logs.
SES-03	Use malformed public key data.	Configuration test fails with invalid-public-key guidance.
SES-04	Repeat connection test.	A new forced SessionKey is requested; existing operational session cache is safely replaceable.
SES-05	Stop Redis and test in development.	In-process cache fallback works; operator receives error only when provider call fails.

Session activation

Some portal configurations may take time before a newly issued session is accepted. LoanHub retries authentication once and preserves uncertain payment attempts for reconciliation instead of generating duplicate customer payments.

7. Borrower C2B single-stage repayment

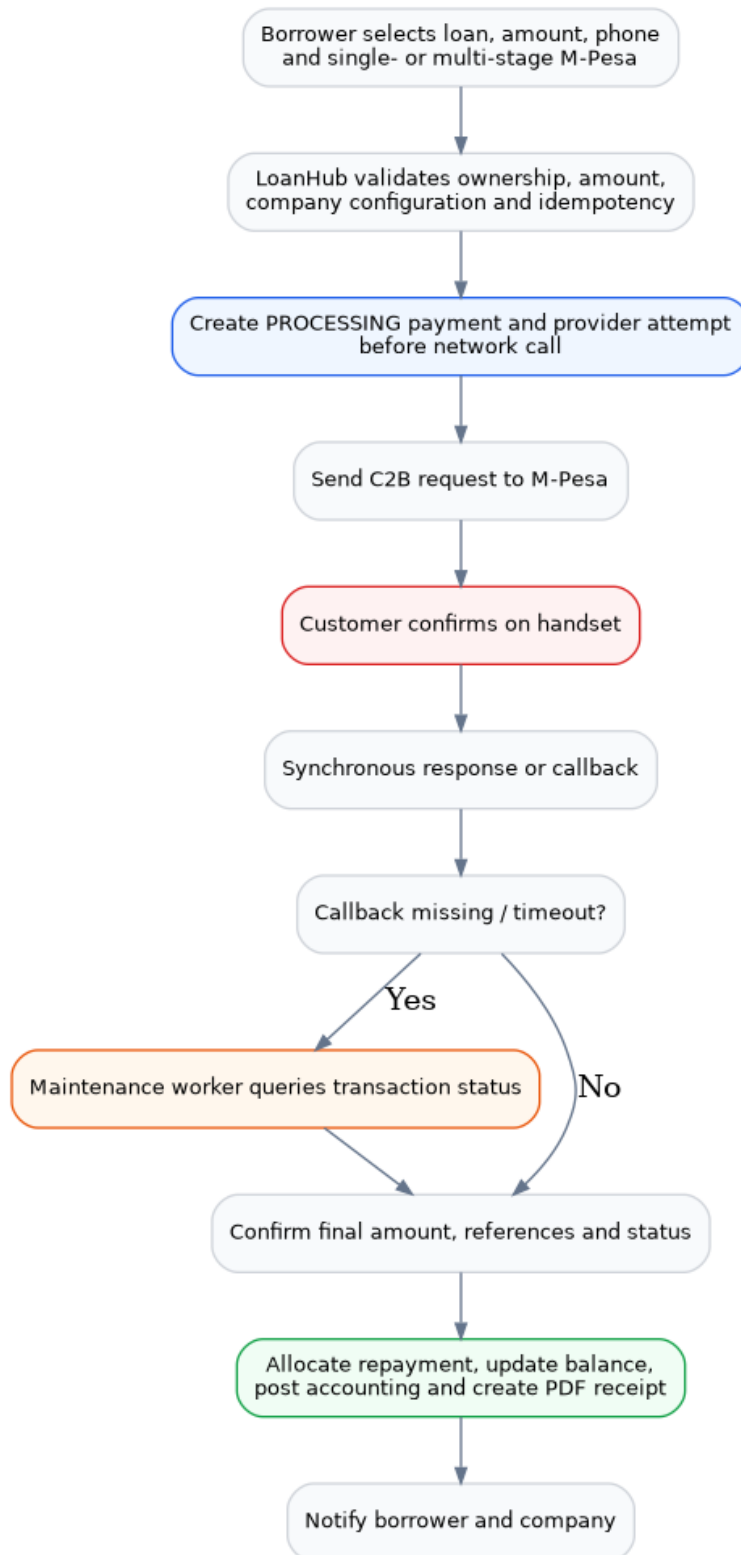


Figure 2 - Borrower repayment, callback and reconciliation flow

7.1 Happy-path test

1. Sign in as a borrower with an active loan.
2. Open Borrower > M-Pesa or the loan repayment action.
3. Select the loan and enter an amount that does not exceed the allowed repayment rules.
4. Enter a valid Lesotho M-Pesa number.
5. Select Single Stage and submit once.
6. Complete the M-Pesa handset prompt using the sandbox scenario.
7. Wait for the callback or select Refresh status.
8. Open the transaction and download the PDF receipt.

Expected checkpoint	Expected value
Local payment	Exactly one PaymentTransaction with provider=mpesa and status processing/succeeded
Provider attempt	Created before the network call with a ThirdPartyConversationID
Final status	succeeded only after conclusive provider result
Loan	Payment allocated and outstanding balance reduced
Accounting	Journal created once
Receipt	One PDF; borrower and authorised company users can download
Notification	Borrower and company receive relevant update

7.2 Idempotency test

Submit the same frontend idempotency key twice. The second request must return the original transaction and must not create another provider attempt or charge request.

7.3 Timeout test

Trigger the provider timeout scenario. The payment and provider attempt must remain processing. Query Transaction Status must run later. Do not submit a second payment to compensate for the timeout.

8. Borrower C2B multi-stage repayment

Multi-stage is available only for loan repayment in this release. It is used when the M-Pesa product requires an additional customer or merchant status step.

Test ID	Procedure	Expected result
C2M-01	Create a repayment and choose Multi Stage.	Operation is c2b_multi; payment is processing after initial acceptance.
C2M-02	Complete customer confirmation.	Callback/status query moves transaction to succeeded when final result is complete.
C2M-03	As authorised finance user, use Update Transaction Status where the provider flow requires it.	Update is recorded against the provider transaction; no direct database-only success is allowed.
C2M-04	Attempt status update on a normal single-stage transaction.	UI does not offer the action or backend rejects inappropriate workflow.
C2M-05	Reject or let the handset prompt expire.	Payment becomes failed/rejected only after provider confirmation.

Field verification

Compare the current Vodacom Lesotho C2B Multi Stage request fields and required update status values with the actual portal application documentation before certification.

9. Callbacks, transaction status and reconciliation

9.1 Callback test

```
POST /api/v1/mpesa/callbacks/transaction-result
Content-Type: application/json

{
  "input_OriginalConversationID": "<provider-conversation>",
  "input_TransactionID": "<mpesa-transaction>",
  "input_ResultCode": "INS-0",
  "input_ResultDesc": "Request processed successfully",
  "input_ThirdPartyConversationID": "<loanhub-reference>",
  "input_OriginalTransactionID": "<original-transaction>",
  "input_Reversed": "False"
}
```

Assertion	Expected result
Acknowledgement	Response code 0 and original/third-party references returned quickly.
Duplicate callback	Same payload hash is acknowledged without duplicate allocation/accounting/receipt.
Unmatched callback	Stored with processing_status=unmatched for platform investigation.
Sensitive data	Callback payload stored through sanitisation; phone/key-like fields are redacted/masked.

9.2 Query Transaction Status

LoanHub queries by M-Pesa TransactionID, ConversationID or its ThirdPartyConversationID. The request uses LES and the company service-provider code. This is the recovery path for timeout, missing callback, closed browser or network interruption.

9.3 Maintenance worker

Checks created/pending/processing provider attempts after a minimum age.

Respects a minimum interval between status queries.

Queries only active company configurations with Query Transaction Status enabled.

Caps status-query attempts and isolates each transaction error.

Applies final payment, loan, accounting and receipt changes exactly once.

10. B2C loan disbursement and beneficiary verification

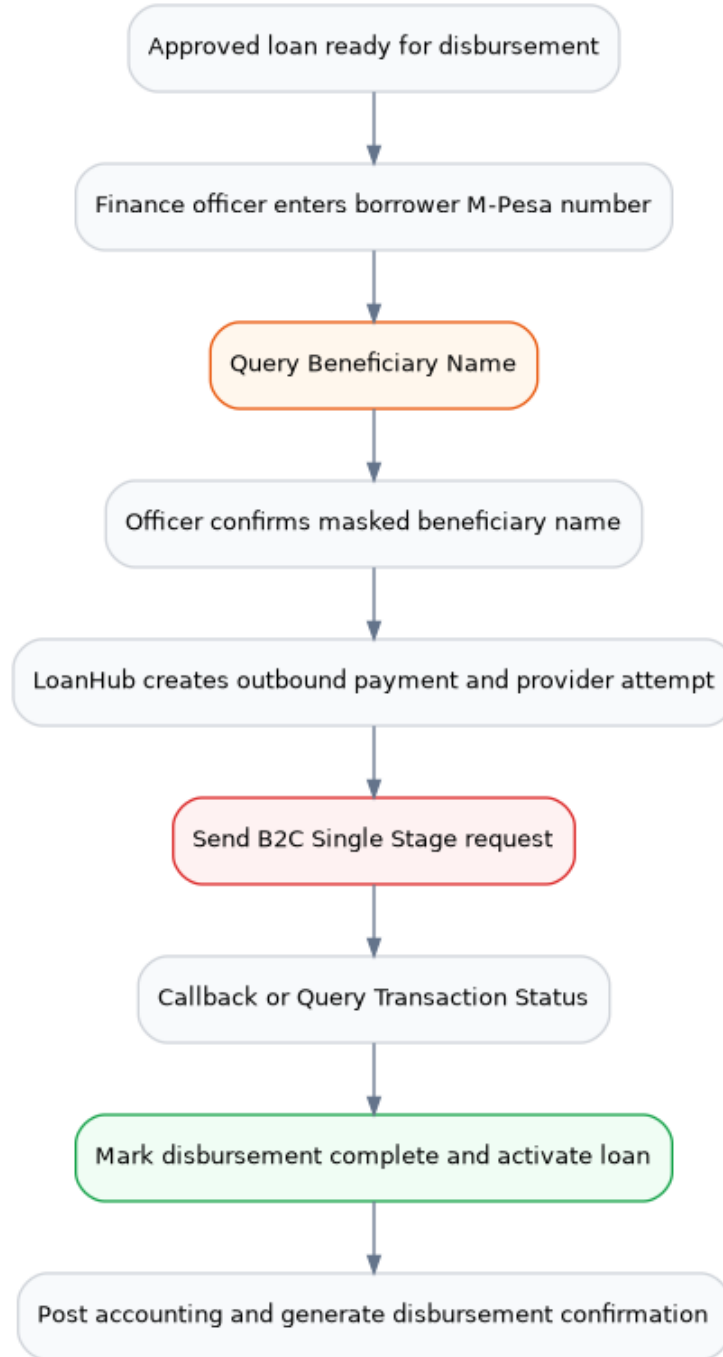


Figure 3 - Beneficiary verification and B2C loan disbursement

1. Approve a walk-in or marketplace loan through normal company controls.
2. Open the company loan record and choose M-Pesa disbursement.
3. Enter the borrower M-Pesa number.
4. Run Query Beneficiary Name.
5. Compare the masked/returned name with the approved borrower identity.

6. Confirm and submit the B2C disbursement.
7. Wait for final provider status or run Refresh status.
8. Confirm the loan is activated only after successful disbursement.
9. Download the disbursement confirmation document.

Negative test	Expected result
Beneficiary name not returned	Disbursement cannot be confirmed as verified.
Wrong company/branch	403 access denial.
Duplicate idempotency key	Original transaction returned.
Timeout	Disbursement remains processing and is reconciled; loan is not activated prematurely.
Provider failure	Loan remains undistributed; safe failure shown.

11. B2B company payment

B2B is intended for approved company payments to a registered business. It is not exposed to ordinary borrowers and requires company-wide finance access.

1. Sign in with a company-wide finance role.
2. Open Company > M-Pesa > Business payments.
3. Enter amount, receiver party code, transaction reference and business-purpose description.
4. Use a unique idempotency key and submit.
5. Confirm provider status by callback or query.
6. Check the outbound accounting journal and transaction confirmation.

Test	Expected result
Branch-only finance user	Blocked from company-level B2B.
Missing receiver party code	Validation error before provider call.
Provider timeout	Payment and provider attempt remain processing and reconcilable.
Completed B2B	Outbound transaction recorded; no loan principal allocation is fabricated.

12. Direct-debit mandate and collection

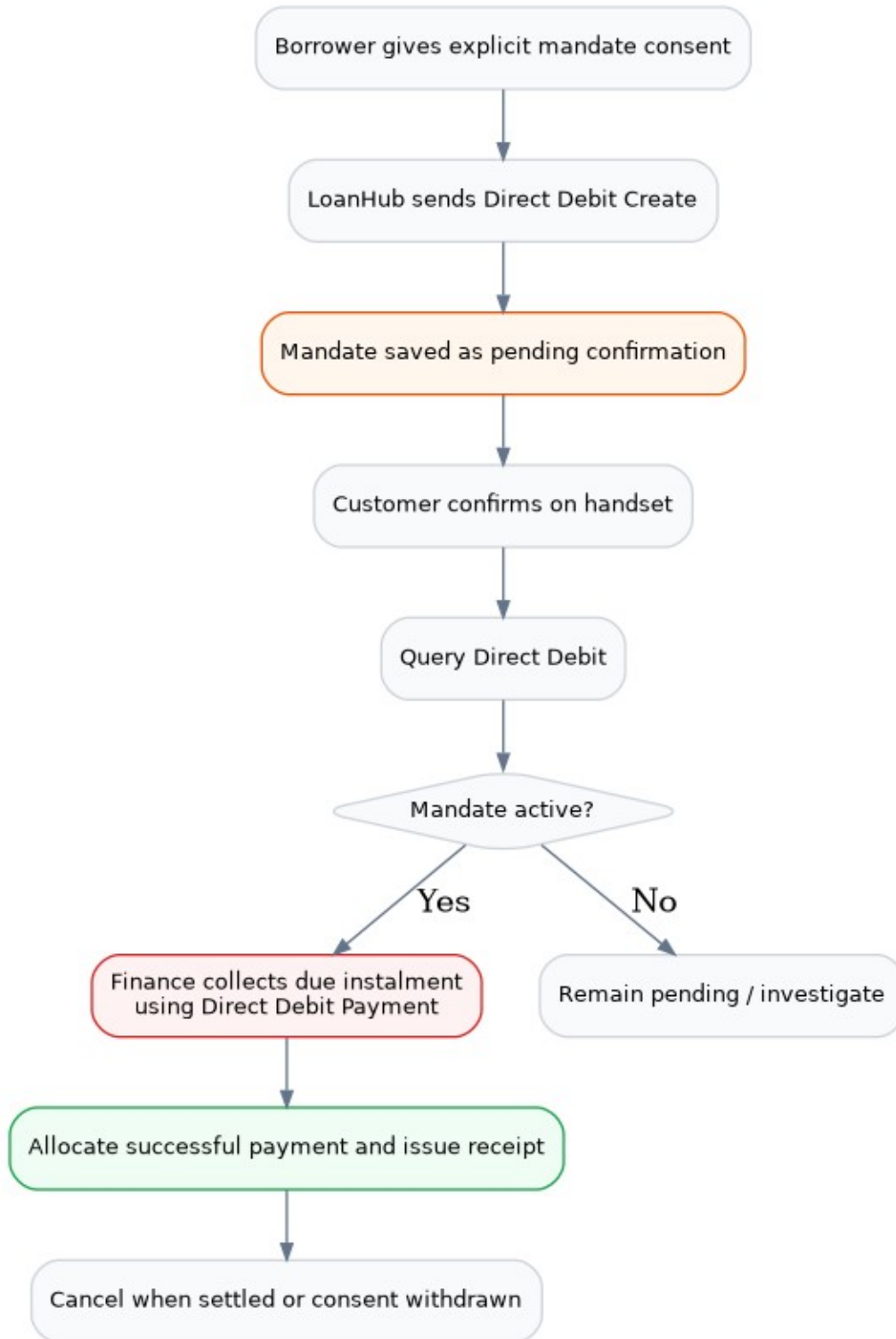


Figure 4 - Direct-debit mandate lifecycle

12.1 Create mandate

1. Open an active borrower loan and choose automatic repayment/direct debit.
2. Enter mobile number, amount or maximum amount, frequency, start/end dates and consent text.
3. Confirm explicit borrower consent.
4. Submit Direct Debit Create.
5. Complete the customer handset confirmation.
6. Run Query Direct Debit until the mandate is active.

Safety rule

A newly created mandate remains pending_confirmation even when the initial request is accepted. LoanHub allows collection only after Query Direct Debit confirms an active state.

12.2 Collect instalment

1. Sign in as authorised finance staff.
2. Open an active mandate and choose Collect.
3. Enter the due amount, transaction reference and unique idempotency key.
4. Submit Direct Debit Payment.
5. Confirm final status, allocation, balance, accounting and receipt.

12.3 Cancel mandate

Borrowers may cancel their own mandate; authorised finance/compliance staff may cancel within company scope. Cancellation is idempotent. Test cancellation when the loan is settled, consent is withdrawn, or the mandate is replaced.

13. Reversal workflow

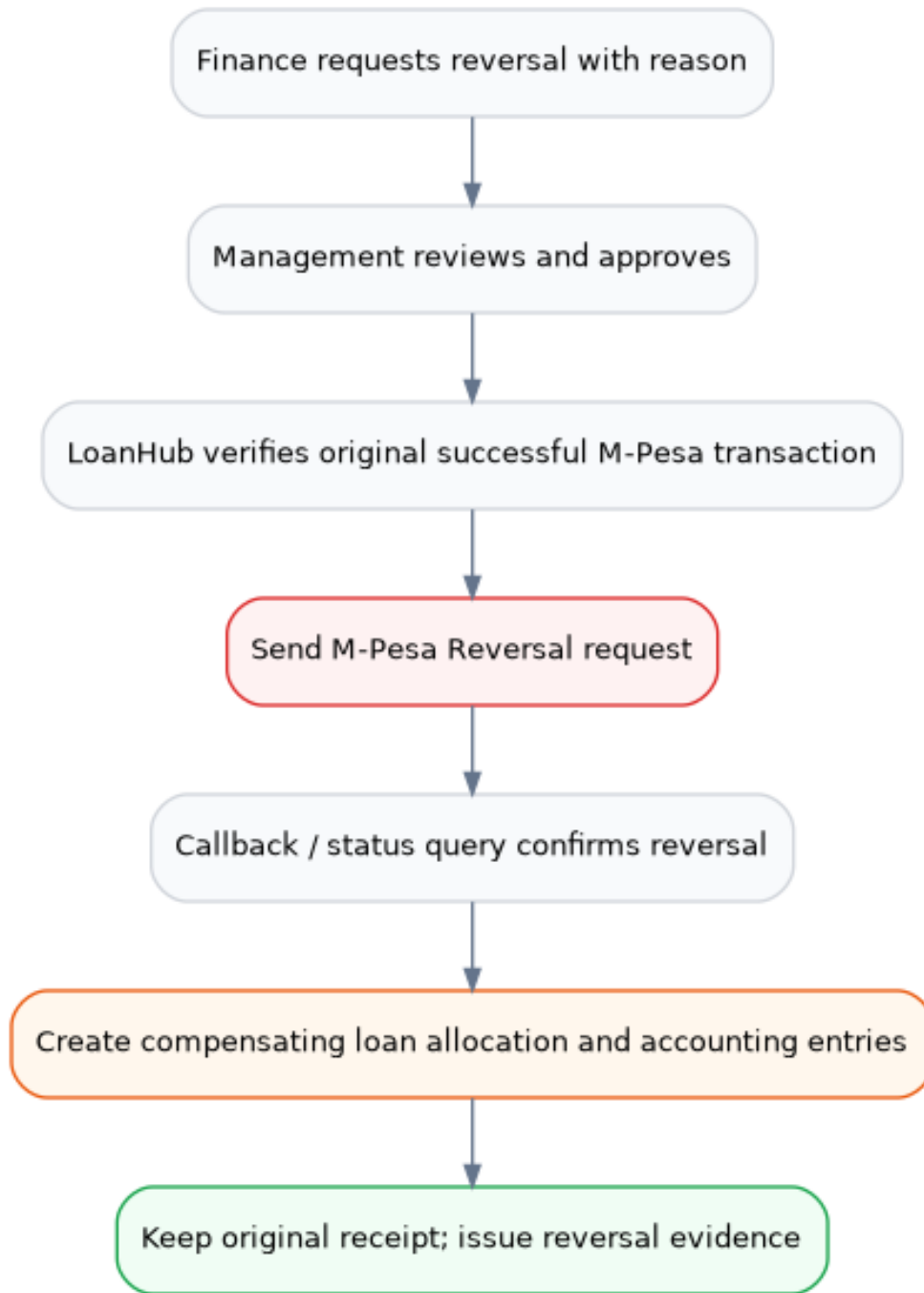


Figure 5 - Controlled M-Pesa reversal workflow

1. Choose a successful M-Pesa transaction.
2. Finance submits a reversal request with a clear reason and amount not greater than the original transaction.
3. Company management reviews and approves.
4. LoanHub verifies that an M-Pesa transaction ID is available.
5. LoanHub sends the provider reversal request.

6. Wait for provider confirmation or query the result.
7. Confirm compensating loan allocation/accounting entries.
8. Confirm the original receipt remains available and reversal evidence is retained.

Negative test	Expected result
Pending/failed original payment	Reversal request rejected.
Amount exceeds original	422 validation error.
Duplicate active request	Existing request returned; no second provider reversal.
No M-Pesa transaction ID	Operator instructed to refresh status first.
Unapproved request	No provider call.

14. Receipts, loan balance and accounting

LoanHub generates a purpose-specific PDF document after confirmed success. The document is stored in the encrypted/managed File Centre and linked to the payment.

Purpose	Document wording	Financial effect
Loan repayment	Repayment receipt	Allocate instalments, reduce balance, post inbound journal.
Loan disbursement	Disbursement confirmation	Activate/disburse loan, post outbound journal.
Refund	Refund confirmation	Outbound transaction and appropriate accounting.
B2B	Business payment confirmation	Outbound business transaction; no loan allocation unless explicitly linked.

Receipt assertions

Document number, M-Pesa reference, date, customer, loan reference, transaction type and method.

Repayment receipt includes balance before, amount, principal, interest, fee, penalty and balance after.

Non-repayment documents use correct labels such as Amount disbursed or Amount transferred.

One verification code and one PDF per payment.

Receipt is unavailable until the payment is conclusively succeeded.

Reversal never deletes the original document.

15. Platform-owner oversight

Open Platform > M-Pesa. The platform owner receives operational visibility without receiving decrypted company credentials.

Metric/list	Test
Configurations	Counts all and active configurations; sandbox and production separated.
Transactions	Totals processing, completed, failed and reversed provider attempts.
Direct-debit mandates	Count matches records across companies.
Callback events	Shows received/unmatched/failed events for support investigation.
Configuration list	Shows metadata and has_api_key/has_public_key flags, never secret values.

Tenant responsibility

The platform owner monitors system health. Each lending company remains responsible for its organisation agreement, provider credentials, settlement, limits, accounting and authorised operational users.

16. User query centre and floating tools

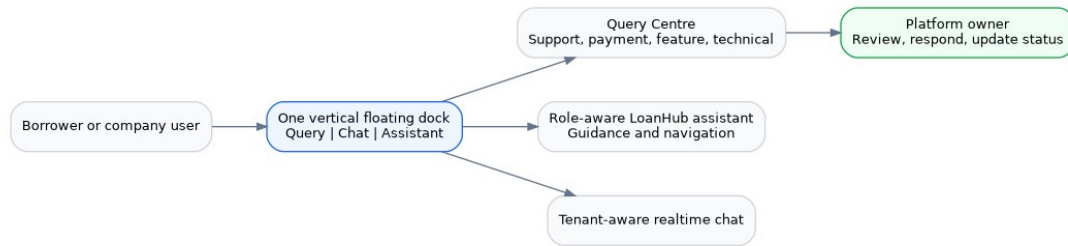


Figure 6 - Query, Chat and Assistant tools

16.1 Submit a query

1. As a borrower or company user, select the Query button in the floating dock.
2. Choose support, payment, feature, technical or training category.
3. Enter title, priority and detailed description.
4. Submit and note the generated query reference.
5. Confirm the query appears in the user history.

16.2 Platform response

1. As platform owner, open Platform > Queries.
2. Filter and open the submitted query.
3. Add the official response and update status.
4. Confirm the submitting user can read the response.

16.3 Floating-tool collision test

Viewport	Expected layout
Phone portrait	Vertical Query, Chat, Assistant dock remains visible; assistant panel opens above entire dock.
Tablet	Dock remains at lower right; panel has responsive height and no action is covered.
Desktop	Assistant panel opens beside the dock; query/chat buttons remain clickable.

17. Error-code and resilience tests

Provider code	Meaning	LoanHub expected action
INS-0	Request processed successfully	Use returned/final status; do not assume settlement from code alone.
INS-1	Internal error	Controlled retry/status enquiry; safe operator message.
INS-6	Transaction failed	Mark failed when conclusive.
INS-9	Request timeout	Keep processing and query original transaction; never duplicate payment.
INS-13	Invalid shortcode	Fail configuration/transaction and alert company administrator.
INS-20	Missing parameters	Developer/configuration defect; no secret leakage.
INS-21	Parameter validation failed	Show safe validation guidance; compare current product docs.
INS-23	Transaction not found	Keep unknown/reconciliation state according to retry policy.
INS-28	Invalid third-party reference	Investigate reference generation and limits.
INS-996	Outside usage time	Retry only during configured usage period.
INS-997	API not enabled	Enable product in M-Pesa application/organisation.
INS-998	Invalid market	Verify vodacomLES and LES.

Resilience scenarios

Kill the browser after submission; callback/worker still completes transaction.

Restart API after creating processing transaction; PostgreSQL provider attempt supports recovery.

Stop callback route temporarily; Query Transaction Status later finalises.

Send the same callback twice; financial side effects happen once.

Simulate Redis outage; no credentials appear and in-process session fallback is bounded.

Simulate database error during callback; event records failed processing for support retry.

18. Security and privacy tests

Test	Expected result
Inspect browser network	No M-Pesa API key, public key, session ID or shortcode secret returned.
Read configuration API response	Only has_api_key/has_public_key booleans; secrets absent.
Inspect logs	MSISDN masked; authorization/key/secret-like fields redacted.
Cross-company access	403 or 404; no foreign transaction data.
Borrower access	Only own borrower transactions/mandates/receipts.
Branch access	Only branch-linked loan transactions.
Callback duplicate	Payload-hash uniqueness prevents duplicate processing.
Production callback URL	HTTPS required.
Production toggle	Cannot select production without explicit production_enabled confirmation.

Credential exposure

Any API key or Bearer session pasted into chat, tickets, screenshots or code should be rotated before further testing.

19. Responsive UI and accessibility tests

Area	Phone	Tablet/Desktop
Borrower repayment dialog	Single-column, scrollable, submit/cancel reachable.	Comfortable width, no excessive stretch.
Company M-Pesa tabs	Scrollable/wrapped navigation; permissions hide forbidden actions.	All applicable tabs readable.
Transaction list	Cards or horizontal-safe table; status visible.	Table/cards use available width.
Floating tools	No overlap with assistant; touch targets at least approximately 44px.	Panel beside dock.
Keyboard	Tab order reaches controls; Escape closes safe dialogs.	Focus visible throughout.
Dark mode	Logo, fields, status chips and text remain readable.	Same.

Minimum viewport checks

320 x 568 phone

390 x 844 modern phone

768 x 1024 tablet

1366 x 768 laptop

1920 x 1080 desktop

20. Regression and go-live sign-off

20.1 Engineering validation

Check	Result / evidence
Python compileall	
M-Pesa unit tests	
Alembic single head	
OpenAPI route generation	
Frontend typecheck	
Frontend lint	
Frontend production build	
Callback public reachability	
Maintenance worker running	
Database backup restore tested	

20.2 Financial user acceptance

Scenario	Tester	Result	Evidence/reference
C2B single repayment			
C2B multi-stage repayment			
B2C disbursement			
B2B payment			
Direct-debit lifecycle			
Reversal			
Receipt and accounting			
Platform monitoring			

20.3 Go-live gate

Developer account linked to the organisation.
 Application sent for organisation review and accepted.
 Vodacom end-to-end testing/certification completed.
 Production products and transaction/customer limits enabled.
 Trusted sources and callback URL registered.
 Production credentials entered through LoanHub and connection tested.
 First transactions monitored and independently reconciled.
 Incident, reversal and support escalation contacts documented.

21. Troubleshooting reference

Symptom	Likely cause	Action
Connection test fails	API key/public key/origin invalid or session not active	Verify portal configuration, rotate exposed keys, retry after provider activation delay.
INS-997	Product not enabled	Enable product in developer application and organisation account.
INS-998	Wrong market	Use vodacomLES and LES.
Payment stuck processing	Callback missing or provider uncertain	Use Refresh status; confirm maintenance worker and Query Status product.
Receipt unavailable	Payment not conclusively succeeded	Refresh/query status; do not create manual success.
Direct debit cannot collect	Mandate pending/not active	Complete handset confirmation and Query Direct Debit.
B2C blocked	Beneficiary not verified or role/branch mismatch	Run QBN and use authorised finance/company scope.
Callback unmatched	References differ from stored attempt	Inspect platform callback events and portal field mapping.
Query/chat/robot overlap	Old frontend bundle or cached CSS	Deploy current frontend, clear .next/browser cache and reload.

Support evidence to collect

LoanHub payment ID and ThirdPartyConversationID.

M-Pesa ConversationID and TransactionID when available.

Company, branch, user role and test timestamp.

Safe response code/description without API keys or SessionKey.

Browser/API logs with personal data masked.

Screenshot of the visible user error and provider sandbox scenario used.

Final reminder

Never mark a transaction succeeded solely to make testing pass. Resolve status through the final provider result or authorised reconciliation process.